Supplement: A Sharp Adaptation Frontier for Heavy-Tailed Bandits with Unknown Moments

This supplement gives complete proofs of all results in the main paper. It also reproves the unknown-scale lower tradeoff with a corrected constant.

1 Algorithm and Exploration Counts

Fix $K \geq 2$ and $\rho \in (0,1)$. Let

$$m_0 = \lceil 16 \log(2K + e) \rceil, \qquad t_0 = K m_0.$$

The algorithm first pulls each arm m_0 times in cyclic order. These pulls are marked as exploration. For $t > t_0$, define

$$D_t = \max\{t_0, \lceil K^{1-\rho} t^\rho \rceil\}.$$

If the exploration count before round t is smaller than D_t, the algorithm explores the next arm in cyclic order. Otherwise it exploits.

Let F_t be the total number of exploration pulls through round t, and let $N_i^E(t)$ be the number assigned to arm i. The initialization has $F_{t_0} = D_{t_0} = t_0$.

Lemma 1 (Exact tracking)**.** *For every $t \geq t_0$, $F_t = D_t$. Moreover,*

$$\left\lfloor \frac{D_t}{K} \right\rfloor \leq N_i^E(t) \leq \left\lceil \frac{D_t}{K} \right\rceil \tag{1}$$

for every arm i.

Proof. Let $g(t) = K^{1-\rho} t^\rho$. For $t \geq K$,

$$g'(t) = \rho K^{1-\rho} t^{\rho - 1} \leq \rho < 1.$$

Hence $g(t) - g(t-1) < 1$ and $\lceil g(t) \rceil - \lceil g(t-1) \rceil \in \{0, 1\}$. Taking the maximum with the constant t_0 preserves this property. Since $g(t_0) = K m_0^\rho \leq K m_0 = t_0$, we have $D_{t_0} = t_0$.

Inductively, suppose $F_{t-1} = D_{t-1}$. If $D_t = D_{t-1}$, the condition for forced exploration is false and $F_t = D_t$. If $D_t = D_{t-1} + 1$, the condition is true, so exactly one exploration pull occurs and again $F_t = D_t$. This proves exact tracking. Cyclic assignment makes the arm counts differ by at most one, which gives (1). ☐

Before any exploitation decision at $t > t_0$, every arm therefore has at least

$$n_t := \lfloor D_{t-1}/K \rfloor \geq \left\lfloor K^{-\rho} (t-1)^\rho \right\rfloor. \tag{2}$$

Since $t_0/K = m_0 \geq 31$, the quantity inside the last floor is at least $m_0^\rho \geq 1$. Since $\lfloor x \rfloor \geq x/2$ for every $x \geq 1$, we have uniformly for $t > t_0$ that

$$n_t \geq \frac{1}{2} K^{-\rho} (t-1)^\rho. \tag{3}$$

The exploration regret admits both uniform and instance-specific bounds. Jensen's inequality implies $\Delta_i \leq 2u^{1/p}$, so

$$R_T^E \leq 2u^{1/p} D_T \leq 2u^{1/p} \{K m_0 + K^{1-\rho} T^\rho + 1\}. \tag{4}$$

On a fixed instance, cyclic assignment and Lemma 1 give

$$\lim_{T \to \infty} \frac{R_T^E}{T^\rho} = K^{-\rho} \sum_{i=1}^{K} \Delta_i. \tag{5}$$

2 Median-of-Means Bounds

We prove the estimator lemma uniformly for all $p \in (1, 2]$. Let $X_1, \ldots, X_n$ be i.i.d., let $\mu = \mathbb{E}X$, and assume $\mathbb{E}|X|^p \leq u$. Let $q \leq n$ be odd and put $k = (q+1)/2$ and $m = \lfloor n/q \rfloor$. Form q block averages $\bar{X}_1, \ldots, \bar{X}_q$ from qm samples and let $M_{n,q}$ be their median.

Lemma 2 (One block)**.** *For every block j,*

$$\mathbb{E}|\bar{X}_j - \mu|^p \leq 16 u (q/n)^{p-1}. \tag{6}$$

Proof. By Jensen, $|\mu|^p \leq u$. Convexity gives

$$\mathbb{E}|X - \mu|^p \leq 2^{p-1} \{\mathbb{E}|X|^p + |\mu|^p\} \leq 2^p u.$$

The von Bahr–Esseen inequality (von Bahr and Esseen, 1965) yields

$$\begin{aligned} \mathbb{E}|\bar{X}_j - \mu|^p &\leq m^{-p}\, 2m \mathbb{E}|X - \mu|^p \\ &\leq 2^{p+1} u m^{1-p}. \end{aligned}$$

Since $n/q \geq 1$, $m = \lfloor n/q \rfloor \geq n/(2q)$. Also $2^{p+1} 2^{p-1} = 2^{2p} \leq 16$. Substitution proves (6). ☐

Lemma 3 (MoM tail and mean)**.** *For every $x > 0$,*

$$\Pr\{|M_{n,q} - \mu| > x\} \le \min\left\{1, \left(\frac{64u(q/n)^{p-1}}{x^p}\right)^k\right\}. \tag{7}$$

If $q \ge 3$, then

$$\mathbb{E}|M_{n,q} - \mu| \le 128u^{1/p}(q/n)^{(p-1)/p}. \tag{8}$$

Proof. By Lemma 2 and Markov's inequality, the probability that one block differs from μ by more than x is at most

$$\theta_x = \min\{1, 16u(q/n)^{p-1}x^{-p}\}.$$

If the median differs by more than x, at least k blocks do. Their independence and a union bound over subsets of size k give

$$\Pr\{|M_{n,q} - \mu| > x\} \le 2^q \theta_x^k.$$

Since $q = 2k - 1$, $2^q \le 4^k$. This proves (7).

Set

$$x_0 = 64^{1/p}u^{1/p}(q/n)^{(p-1)/p}.$$

Integrating (7) gives

$$\begin{aligned}\mathbb{E}|M_{n,q} - \mu| &\le x_0 + \int_{x_0}^{\infty} (x_0/x)^{pk}\, dx \\ &= x_0\left(1 + \frac{1}{pk-1}\right).\end{aligned}$$

For $q \ge 3$, $k \ge 2$ and $pk-1 > 1$. Hence the last display is at most $2x_0 \le 128u^{1/p}(q/n)^{(p-1)/p}$. ◻

Lemma 4 (Maximum over arms)**.** *Suppose K arm estimators satisfy* (7) *with arm-specific q_i, n_i, k_i, where $q_i \le q$, $n_i \ge n$, and $k_i \ge \log(2K)$. Then*

$$\mathbb{E}\max_{i \le K} |M_i - \mu_i| \le 128e\, u^{1/p}(q/n)^{(p-1)/p}. \tag{9}$$

No independence across arms is required.

Proof. Let $k_0 = \min_i k_i$ and $A = 64u(q/n)^{p-1}$. The union bound and (7) imply

$$\Pr\{\max_i |M_i - \mu_i| > x\} \le \min\{1, K(A/x^p)^{k_0}\}$$

whenever $x^p \ge A$. Let $x_0 = A^{1/p}K^{1/(pk_0)}$. Tail integration as in the previous proof gives an expectation at most $2x_0$. Since $k_0 \ge \log(2K)$,

$$K^{1/(pk_0)} \le \exp\{\log K/\log(2K)\} \le e.$$

Also $2A^{1/p} \le 128u^{1/p}(q/n)^{(p-1)/p}$. ◻

For the algorithm, let $\ell : \mathbb{N} \to [1, \infty)$ be nondecreasing with $\ell(n) \to \infty$, $\ell(n+1) - \ell(n) \le 1$, and $\log \ell(n) = o(\log n)$. Define

$$\begin{aligned}q(n) = \max\{r : r \le n,\ & r \le 2\lceil \log(2K) + \ell(n)\rceil + 1, \\ & r \text{ is odd}\}.\end{aligned}$$

The assumed schedule is nondecreasing and satisfies $\ell(n+1) - \ell(n) \le 1$. Hence $q(n+1) - q(n) \le 2$. This is the only regularity property of the block schedule needed below. For $n \ge m_0$, both arguments of the minimum exceed $2\log(2K)+1$. Thus $q(n) \ge 2\log(2K)-1$, and $k(n) = (q(n)+1)/2 \ge \log(2K)$. Also

$$q(n) \le 3 + 2\log(2K) + 2\ell(n). \tag{10}$$

3 Proof of the Finite-Time Upper Bound

Let J_t be the arm chosen on an exploitation round and fix an optimal arm i_*. Since $M_{J_t} \ge M_{i_*}$,

$$\begin{aligned}\Delta_{J_t} &= \mu_* - \mu_{J_t} \\ &\le |M_{i_*} - \mu_*| + |M_{J_t} - \mu_{J_t}| \\ &\le 2\max_i |M_i - \mu_i|. \end{aligned} \tag{11}$$

The exploration counts of any two arms differ by at most one. Applying Lemma 4, (3), and (10), we obtain for all exploitation rounds outside a finite initial interval

$$\begin{aligned}\mathbb{E}\Delta_{J_t} &\le Cu^{1/p}\left(\frac{1 + \log(2K) + \ell(t)}{K^{-\rho}t^{\rho}}\right)^{(p-1)/p} \\ &= Cu^{1/p}K^{\rho(p-1)/p}L_t^{(p-1)/p}t^{-\rho(p-1)/p}, \end{aligned} \tag{12}$$

where $L_t = 1 + \log(2K) + \ell(t)$.

Put $a = \rho(p-1)/p$. Since $p \le 2$ and $\rho < 1$, $a < 1/2$. Monotonicity of L_t and integral comparison give

$$\sum_{t=1}^{T} L_t^{(p-1)/p}t^{-a} \le 2L_T^{(p-1)/p}T^{1-a}. \tag{13}$$

Combining (4), (12), and (13), and enlarging the universal constant to cover $T \le t_0$, proves

$$\begin{aligned}R_T \le Cu^{1/p}\{&K\log(eK) + K^{1-\rho}T^{\rho} \\ &+ K^{\rho(p-1)/p}T^{1-\rho(p-1)/p}L_T^{(p-1)/p}\}.\end{aligned}$$

This is Theorem 2 of the main paper.

If $\rho \le 2/3$, then for every $p \in (1, 2]$,

$$1 - \rho(p-1)/p - \rho = 1 - \rho(2p-1)/p \ge 1 - 3\rho/2 \ge 0.$$

Thus the exploitation term has the larger or equal exponent.

4 Proof of the Fixed-Instance Bound

Fix an instance in $\mathcal{H}_{p,u}$ and a suboptimal arm i with $\Delta_i > 0$. If i is selected on an exploitation round, then

$$M_i \geq M_{i_*}.$$

This implies either $M_i - \mu_i \geq \Delta_i/2$ or $\mu_* - M_{i_*} \geq \Delta_i/2$. Let $\underline{q}_t$ and $\overline{q}_t$ be the smaller and larger block counts among the two estimates. The arm exploration counts differ by at most one, so $\overline{q}_t \leq \underline{q}_t + 2$. By Lemma 3, for a constant C and all sufficiently large t,

$$\Pr\{J_t = i\} \leq 2 \left[\frac{Cu\overline{q}_t^{p-1}}{\Delta_i^p n_t^{p-1}} \right]^{(\underline{q}_t+1)/2}. \tag{14}$$

Here n_t is the smaller sample count.

We verify summability. Since $\log \ell(n) = o(\log n)$,

$$\log \overline{q}_t = o(\log n_t).$$

Hence, for every fixed p, u, Δ_i, the logarithm of the quantity inside the brackets in (14) is eventually at most

$$-\frac{p-1}{2} \log n_t.$$

By (3), it is then at most $-(p-1)\rho \log t/4$ after increasing the instance-dependent threshold. Since $\underline{q}_t \to \infty$, eventually

$$\frac{(p-1)\rho(\underline{q}_t + 1)}{8} \geq 3.$$

Therefore $\Pr\{J_t = i\} \leq 2t^{-3}$ eventually. Summing over time and over the finitely many suboptimal arms proves

$$\sum_{t=1}^{\infty} \Pr\{t \text{ is exploit and } J_t \text{ is suboptimal}\} < \infty.$$

The expected total regret due to exploitation is finite. Equation (5) now proves

$$\limsup_{T\to\infty} \frac{R_T}{T^\rho} \leq K^{-\rho} \sum_i \Delta_i.$$

5 The Unknown-Scale Lower Tradeoff

For completeness, we prove the exponent lower bound used in the main paper. The argument follows Genalti and Metelli (2025), which adapts Hadiji and Stoltz (2023). Our statement uses a p-dependent constant. This is all that is needed for the polynomial frontier.

Theorem 5 (Lower tradeoff)**.** *Fix $p \in (1, 2]$. Suppose a policy does not know u and, for a function $\Phi(T) = o(T)$, satisfies*

$$R_T(\eta) \leq v^{1/p}\Phi(T) \tag{15}$$

for every $v > 0$, every $\eta \in \mathcal{H}_{p,v}$, and every T. Then every fixed instance ν with at least one suboptimal arm satisfies

$$\liminf_{T\to\infty} \frac{R_T(\nu)}{(T/\Phi(T))^{p/(p-1)}} \geq c_p \sum_{i:\Delta_i>0} \Delta_i, \tag{16}$$

where one may take

$$c_p = \frac{1}{2\, 8^{p/(p-1)}}. \tag{17}$$

Proof. Fix a suboptimal arm a of ν, with law ν_a, mean μ_a, and gap $\Delta_a > 0$. Let v be any finite common upper bound on the arms' pth absolute moments. For $\beta \in (0, 1/2)$, define an alternative law

$$\nu'_{a,\beta} = (1-\beta)\nu_a + \beta\delta_{\mu_a + 2\Delta_a/\beta}. \tag{18}$$

Its mean is $\mu_a + 2\Delta_a = \mu_* + \Delta_a$, so arm a is uniquely better than every original optimal arm by at least Δ_a. Let v'_β be a common pth-moment bound for the alternative bandit. We may take

$$v'_\beta = v + (1-\beta)\mathbb{E}_{\nu_a}|X|^p + \beta|\mu_a + 2\Delta_a/\beta|^p. \tag{19}$$

The redundant v term safely covers all unchanged arms.

Write $N_a(T)$ for the number of pulls of arm a, and let P and P' be the history laws under the original and alternative instances. Mixture domination gives

$$\mathrm{KL}(\nu_a, \nu'_{a,\beta}) \leq \log \frac{1}{1-\beta} \leq 2\beta \log 2.$$

The standard bandit change-of-measure identity and data processing give

$$\mathrm{kl}\left(\frac{\mathbb{E}_P N_a(T)}{T}, \frac{\mathbb{E}_{P'} N_a(T)}{T} \right) \leq 2\beta \log 2\, \mathbb{E}_P N_a(T). \tag{20}$$

For $x, y \in [0, 1]$, binary relative entropy obeys

$$\mathrm{kl}(x, y) \geq (1-x) \log \frac{1}{1-y} - \log 2. \tag{21}$$

The moment-free guarantee implies

$$\mathbb{E}_P N_a(T) \leq \frac{v^{1/p}\Phi(T)}{\Delta_a}, \tag{22}$$

$$T - \mathbb{E}_{P'} N_a(T) \leq \frac{(v'_\beta)^{1/p}\Phi(T)}{\Delta_a}. \tag{23}$$

Substituting (22) and (23) into (20) through (21) yields

$$\left(1 - \frac{v^{1/p}\Phi(T)}{T\Delta_a}\right) \log\left(\frac{T\Delta_a}{(v'_\beta)^{1/p}\Phi(T)} \right) - \log 2 \leq 2\beta \log 2\, \mathbb{E}_P N_a(T). \tag{24}$$

Put $r = p/(p-1)$ and choose

$$\alpha = 8^{-r}, \qquad \beta_T = \alpha^{-1}\left(\frac{\Phi(T)}{T}\right)^r. \tag{25}$$

Since $\Phi(T) = o(T)$, $\beta_T \to 0$. Equation (19) gives

$$(v'_{\beta_T})^{1/p}\Phi(T) = 2\Delta_a \alpha^{(p-1)/p} T\{1 + o(1)\}. \tag{26}$$

The right side equals $\Delta_a T\{1+o(1)\}/4$. The first factor on the left of (24) tends to one. The logarithm tends to $\log 4$, so the entire left side tends to $\log 2$. Using (25) on the right gives

$$\begin{aligned}\liminf_{T\to\infty} \frac{\mathbb{E}_P N_a(T)}{(T/\Phi(T))^r} &\geq \frac{\alpha}{2\log 2}\log 2 \\ &= \frac{1}{2\,8^r} = c_p.\end{aligned}$$

Multiplying by Δ_a and summing this inequality over all suboptimal arms proves (16). □

If $\Phi(T) = T^{a+o(1)}$ and a fixed-instance rate is $T^{b+o(1)}$, Theorem 5 implies

$$b \geq \frac{p}{p-1}(1-a),$$

or equivalently

$$b + \frac{p}{p-1}a \geq \frac{p}{p-1}.$$

For the algorithmic exponent $a = 1 - \rho(p-1)/p$, the right side is exactly ρ. This completes the proof of the simultaneous sharp frontier.

References

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